Apache Spark: Aiding Big Data Cluster Computing



Apache Spark is a unified analytics engine for Big Data processing, and has support for both batch and stream processing. What is it that makes it an attractive solution for developers and data scientists alike?

Data is everywhere and data is everything. Every purchase at a supermarket generates data. Every post on Twitter generates data. Every administered vaccine generates data. Every flight that takes off generates a stream of data.

These examples are sufficient to say that enormous amounts of data are getting generated every moment across the world. Organisations tend to look into this data for insights, predictions, analytics, and so forth. However, the data is now so big in size that traditional technologies are incapable of processing it. This is where Big Data steps in.

Big Data is a set of tools and technologies to process huge amounts of data. More and more organisations across various verticals have been implementing Big Data for their business needs in the last few years.

Apache Spark is one of the most promising open source unified engines to process Big Data. This article introduces the length and breadth of Apache Spark to help CTOs make architectural decisions for their organisations.

Architecture

Apache Spark is a cluster computing system that processes data in memory.

The cluster consists of a master node and a minimum of one worker node. In practice, the cluster usually consists of scores of worker nodes. Though Spark can manage the cluster on its own, other time-tested tools like Hadoop Yarn, Apache Mesos, or even Kubernetes can also be used for managing the cluster for Spark.

How does Spark use the cluster for computing?

Let's take a simple analogy. Assume that a CEO wants to shortlist a few candidates for an important position in the organisation. Thousands of resumes are received in response to an advertisement. It takes significant time and effort for a single person to filter them based on the parameters like